

Effects of coffee husk biochar on the properties of concrete

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Highlights

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Holistic study of CHB as a cement replacement in concrete, before and after accelerated aging.

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CHB produced through [pyrolysis](#) at 450°C was added at 1 %, 3 %, and 5 % cement replacement levels.

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1 % CHB reduced density and [thermal conductivity](#) while preserving strength of concrete.

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CHB's porosity acted as an internal water reservoir, sustaining [cement hydration](#) during curing.

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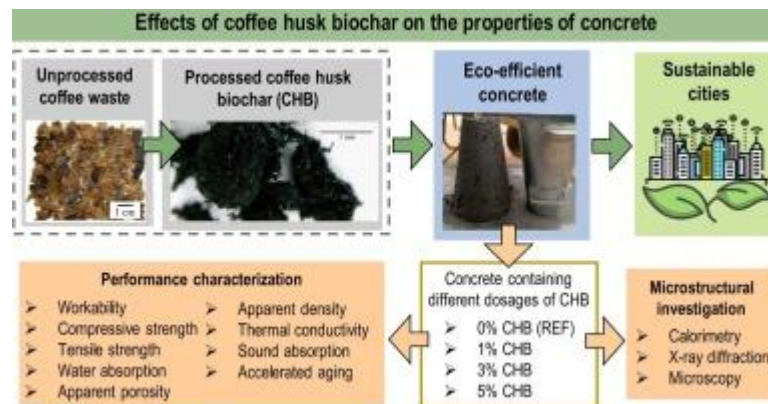
CHB incorporation did not affect statistically the [acoustic performance](#) of concrete.

Abstract

The incorporation of different types of biochar into cement-based materials can enhance sustainability by recycling agro-industrial wastes and potentially reducing the environmental impact of [cement production](#). However, previous research did not report the effects of coffee husk biochar (CHB) on mechanical, rheological, thermal, acoustic, durability, mineralogical and morphological [properties of concrete](#). In this study, the CHB

was produced through the [pyrolysis](#) of coffee husks at 450°C and used as a cement replacement at 1 %, 3 %, and 5 % (by mass). The CHB was characterized in terms of unit mass, [particle size distribution](#), water absorption, and microstructure. The cement-based composites were subjected to workability, [calorimetry](#), [compressive strength](#), [tensile strength](#), water absorption, [apparent porosity](#), apparent density, [thermal conductivity](#), [sound absorption](#), X-ray diffraction, microscopy, and accelerated aging tests. [Concrete specimens](#) with 1 % CHB demonstrated lower density and [thermal conductivity](#) than the reference specimens, while still maintaining adequate [mechanical strength](#). The [cement hydration reactions](#) were slightly inhibited, and water was gradually released from the CHB's pores. Moreover, the incorporation of CHB had no impact on the [acoustic performance](#) of the concrete. As a result, 1 % CHB was identified as the optimal replacement percentage for concrete.

Graphical Abstract



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Introduction

Concrete is the most widely used material in the construction industry worldwide, employed in structures ranging from simple buildings to highly complex and large-scale projects, thanks to its moldability in the fresh state and high strength once it has hardened [1], [2]. Concrete is composed of cement, water, fine and coarse aggregates, with aggregates playing a key role in influencing its physical and chemical properties, while admixtures can be incorporated to modify its basic characteristics [3], [4], [5].

Biochar, a carbon-rich material with low density, high porosity, and good water absorption, is produced through the pyrolysis process, in which biomass is thermochemically decomposed at temperatures between 300 °C and 1300 °C in an inert atmosphere [6], [7], [8]. In fact, Jabar [9] state that conventional pyrolysis is a thermochemical process that

decomposes biomass at relatively low heating rates and moderate temperatures. Due to longer vapor retention time of condensable and non-condensable gaseous fractions, the condensable volatile organic fractions react with one another to form high quantity of solid fraction. Due to these conditions, conventional pyrolysis is particularly effective for production of different types of biochar, using various biomass sources such as wood, rice husks, sugarcane bagasse, peanut shells, food waste, coffee husks, and others [10], [11], [12], [13], [14].

In 2024, global coffee production reached approximately 10 million tons [15]. Approximately 50 % of the volume produced during the coffee peeling process consists of coffee husks [16]. Hence, coffee cultivation generates large volumes of agro-industrial wastes, which require proper disposal to avoid environmental issues and to provide economic benefits [17], [18]. In fact, agricultural and forestry practices are the main contributors to the substantial production of biomass waste. Around 140 Gt of waste are generated globally each year, creating significant management challenges, as discarded biomass has a negative environmental impact [19].

Previous research has demonstrated the positive effects of varying biochar contents and types incorporated into cement-based matrices, including improvements in cement hydration, contaminant immobilization, fracture energy, and resistance to high temperatures, along with both direct and indirect influences on mechanical strength [20], [21], [22], [23], [24].

The experimental study developed by Gupta, Tulliani, and Kua [25] demonstrated that biochar particle size significantly affects the workability, hydration, and mechanical properties of cementitious matrices. The authors investigated the use of biochar prepared from waste coconut shell (COCO-BC) and mixed tropical wood (WBC). Coarser biochar (200–300 μm) increased water absorption and accelerated hydration and setting times compared to finer biochar (50–80 μm). A blend of coarser and finer biochar particles balanced these effects, improving packing density and enhancing compressive strength at early ages, although strength gains were limited by macro-pores in coarser particles. Moreover, blending particle sizes reduced biochar processing energy consumption by 27–37 %, which can provide savings of energy cost

Gupta et al. [26] evaluated the potential of biochar from coconut and wood waste to partially replace conventional binders in mortar mixtures, focusing on hydration, mechanical, permeability, and shrinkage behavior. The use of small contents of biochar enhanced early hydration through filler effects and increased peak heat release. Thermal treatment of coconut shell biochar significantly improved surface area and porosity, resulting in a 20–30 % increase in compressive strength compared to untreated biochar.

The combination of biochar (5 %, by weight of cement) and silica fume (10 %, by weight of cement) reduced autogenous shrinkage and drying shrinkage. Biochar addition also reduced overall and initial water absorption, particularly when replacing cement, though silica fume substitution had minimal impact on total absorption. These results suggested biochar can serve as a sustainable, performance-enhancing alternative to silica fume and cement in mortars.

According to a comprehensive review article on biochar-modified concrete published by Murali and Wong [27], fresh concrete containing biochar showed densities between 2245 and 2330 kg/m³. After hardening, the density values decreased, ranging from 2013 to 2195 kg/m³. The tensile strength varied from 1.9 to 4.9 MPa, depending on biochar content between 0.1 % and 5.0 %. Similarly, flexural strength ranged from 2.1 to 8.8 MPa, influenced by biochar concentrations from 0.25 % to 10 %.

Coffee husk is a potassium-rich agricultural residue that is abundantly generated during the processing of mature coffee fruits. Despite its high nutrient content, coffee husk remains largely underutilized by farmers, primarily due to logistical challenges in transporting and applying it in its raw form to plantations [28]. This underuse poses environmental burdens and represents a missed opportunity for recycling these types of residues. The transformation of this waste into CHB provides a scientifically grounded pathway for value addition, particularly in eco-friendly cementitious materials. Despite this, no studies have yet investigated the effects of different concentrations of coffee husk biochar (CHB), derived from the pyrolysis of coffee husks, on the mechanical, rheological, thermal, acoustic, durability, and microstructural properties of concrete. Understanding these effects is essential for assessing the feasibility of using CHB as a sustainable admixture to produce eco-efficient cementitious composites and optimize its potential benefits in enhancing concrete properties.

To narrow this knowledge gap, the present study offers the following original contributions:

(i) design of concrete incorporating CHB derived from the husks of mature coffee fruits, ensuring compliance with standard requirements for mechanical strength development; (ii) new insights related to the promising use of CHB as a lightweight and porous admixture serving as a water reservoir to sustain cement hydration during the curing period; (iii) pioneering elucidation of quantitative effects of different replacement levels of cement by CHB on the mechanical, rheological, thermal, acoustic and durability properties of concrete; and (iv) comprehensive mineralogical and morphological analyses through the combination of calorimetry tests, microscopy analyses, and X-ray diffraction (XRD) techniques.

Materials

The coffee husks used for production of CHB were agro-industrial wastes generated from Inovacafé, located at the Universidade Federal de Lavras (Lavras, Minas Gerais, Brazil). The husks were obtained through a dry peeling process of beans of *Coffea arabica*. The concrete specimens were produced with Portland cement CII F-32, medium sand (fine aggregates), and crushed stone (coarse aggregates), all obtained from local suppliers in the city of Lavras (Minas Gerais, Brazil).

Production of CHB

The CHB was obtained

Biochar yield

Under pyrolysis at 450 °C, a CHB yield of 32.27 % was verified. In a similar process, Veiga et al. [46] obtained a biochar yield of approximately 40 %. This difference can be attributed to variations in operational parameters, such as heating rate and holding time at the maximum temperature. The biochar yield observed in this work is consistent with findings of Bridgwater [47], which observed an average biochar yield of 35 % for various biomasses when pyrolyzed at temperatures between 300 °C

Conclusions

Therefore, the present work contributed to the current state-of-the-art by investigating, for the first time, the effects of CHB on the mechanical, rheological, thermal, acoustic, durability, mineralogical and morphological properties of concrete. Unlike previous works that explored other biochar sources, this study examines CHB produced at 450°C and incorporated at 1 %, 3 %, and 5 % as a cement replacement, providing the following insights into its potential for sustainable concrete production:

CRedit authorship contribution statement

Nayane Pereira Freire Coelho: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gustavo Henrique Nalon:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Juliana Farinassi Mendes:** Methodology, Investigation, Formal analysis, Data curation. **Tiago José Pires de Oliveira:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rafael Farinassi**

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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